CS 225

#9: Linked Memory

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Class templates:

List.h

```
1  #ifndef LIST_H
2  #define LIST_H
3
4  template <class T>
5  class List {
6    public:
7
8
9
10
11
12
13
14
15
16
17
18   private:
19
20 };
21
22 #endif
```

Two Basic Implementations:

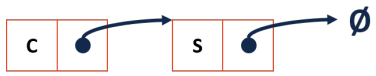
1.

2.

Linked Memory:

Snippet: ListNode struct

```
1  struct ListNode {
2    T data;
3    ListNode * next;
4    ListNode(T & data) : data(data), next(nullptr) { }
5  };
```



Coding with Linked Lists: Examples

List.h

```
 1  #ifndef LIST_H
 2  #define LIST_H
 3
 4  template <class T>
 5  class List {
 6    public:
…        /* ... */
28    private:
29      struct ListNode {
30        T data;
31        ListNode * next;
32        ListNode(T & data) : data(data), next(nullptr) { }
33      };
34
35
36
37
38
39
40  };
41
42  #include "List.cpp"
43
44  #endif
```

List.cpp

```
 1  template <class T>
 2  void List<T>::insertAtFront(const T& t) {
 3
 4
 5
 6
 7
 8
 9
10
11
12  }
```

List.cpp

```
 1  template <class T>
 2  void List<T>::printReverse() const {
 3
 4
 5
 6
 7
 8
 9
10
11
12  }
```

List.cpp

```
 1  template <class T>
 2  void List<T>::print_(ListNode *cur) const {
 3
 4
 5
 6
 7
 8
 9
10
11
12  }
```

CS 225 – Things To Be Doing:

1. Lab 2 is due March 8
2. MP2 is due March 16
3. MP2 Extra Credit w/ Part 1 due March 9